

Hands On-1-Spring Data JPA - Quick Example

OrmLearnApplication.java

```
package com.cognizant.ormlearn;

import com.cognizant.ormlearn.service.CountryService;

import java.util.List;

import org.slf4j.Logger;

import org.slf4j.LoggerFactory;

import org.springframework.boot.SpringApplication;

import
org.springframework.boot.autoconfigure.SpringBootApplication;

import org.springframework.context.ApplicationContext;

import com.cognizant.ormlearn.model.Country;

@SpringBootApplication(scanBasePackages =
"com.cognizant.ormlearn")

public class OrmLearnApplication {

    private static final Logger LOGGER =

        LoggerFactory.getLogger(OrmLearnApplication.class);

    private static CountryService countryService;

    public static void main(String[] args) {

        ApplicationContext context =

            SpringApplication.run(OrmLearnApplication.class,args);

        countryService=context.getBean(CountryService.class);

        LOGGER.info("Inside Main");
```

```

        testGetAllCountries();
    }

    private static void testGetAllCountries(){

        LOGGER.info("Start");

        List<Country> countries=

            countryService.getAllCountries();

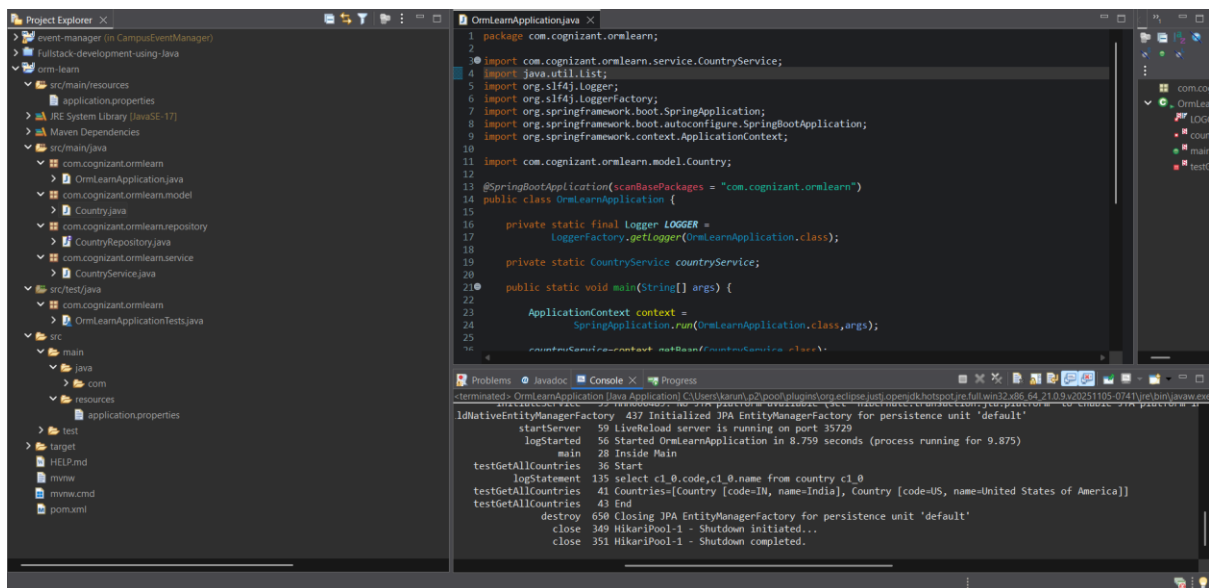
        LOGGER.debug("Countries={}",countries);

        LOGGER.info("End");

    }

}

```



Hands On-4- Difference between JPA, Hibernate and Spring Data JPA

Java Persistence API (JPA)

What is JPA?

- JPA stands for Java Persistence API.
- It is a Specification (JSR 338).
- It defines rules for storing Java objects in a relational database.
- It is not an implementation.

Features of JPA

- Object Relational Mapping (ORM)
- Entity Management
- Relationships
- JPQL
- Transactions

Hibernate

What is Hibernate?

Hibernate is an ORM framework.

It implements JPA.

Hibernate converts Java objects into SQL statements.

Hibernate Responsibilities

- Generates SQL
- Opens database connection
- Executes SQL
- Converts ResultSet into Objects
- Caching
- Transactions

Spring Data JPA

What is Spring Data JPA?

Spring Data JPA is not another ORM framework.

It sits on top of Hibernate and removes boilerplate code.

It automatically creates Repository implementations.

Why is Spring Data JPA easier?

Hibernate needs

- Session
- Transaction
- Commit
- Rollback
- Close Session

Spring Data JPA automatically manages all these.

Relationship

JPA

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Rules

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Hibernate

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Implements JPA

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Spring Data JPA

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Uses Hibernate

Reduces code

Auto CRUD

Repository Support

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Database